

Ahmed Mohammed Abdelgelel

Software Engineer | Co-Founder @ Penta Studio

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PROFILE

Software Engineer with 2 years of experience in Node.js and JavaScript, building scalable and production-ready systems. Experienced in designing RESTful APIs, managing sessions and authentication flows, and structuring backend architectures for real-world applications. Worked on multiple client projects, focusing on performance, clean architecture, and maintainability.

EXPERIENCE

Co-Founder & Software Engineer | Penta Studio – Remote

Dec 2025 – Present

- Co-founded a digital product studio delivering secure, scalable web and mobile solutions for startups and enterprises.
- Architected and led backend development for client projects using Node.js frameworks such as ExpressJS and NestJS, and cloud infrastructure.
- Oversaw system design, API architecture, and third-party integrations across multiple simultaneous projects.
- Collaborated with frontend team, managed client communication, and drove product from concept to launch.

Backend Developer | Amen Stories

Feb 2025 – Present

- Built and shipped a full backend for a story-commerce platform supporting end-to-end order flows — from cart and Stripe payments to automated EPUB generation and artist assignment workflows.
- Implemented artist assignment workflows, multilingual story structures, and media upload pipelines for dynamic content management.
- Engineered personalized EPUB generation with automated background jobs for order processing and session maintenance.
- Designed role-based access for admins, artists, and customers, ensuring scalable data modeling for story customization.

Naqaa Al-Ain – Car Maintenance System | Freelance Backend Developer – Saudi Arabia

Apr 2025 – Aug 2025

- Designed and implemented RESTful APIs for orders, invoices, and payment workflows.
- Built role-based authentication and authorization system for admins, staff, and clients.
- Integrated Azure OCR to auto-extract vendor, date, and amount data from uploaded receipts.
- Optimized database workflows to ensure reliability and scalability under production load.

PROJECTS

Smart Cache Engine – Redis-like In-Memory Cache

- Built a high-performance Redis-inspired cache engine from scratch in Node.js achieving 3M+ ops/sec throughput.
- Implemented O(1) LRU eviction via a doubly linked list + hashmap, dual-layer TTL expiration (lazy on read + active background sweep), and byte-level memory cap enforcement.
- Added Redis-style Sorted Sets with lazy-sort optimization, Stampede Guard (eliminates thundering herd — 99.9% fewer backend calls), and p50/p95/p99 latency metrics via reservoir sampling.
- Validated with 79 unit tests (zero external dependencies) across 5 modules; benchmarked across 10 load scenarios.

EPUB-to-PDF Converter

- Unpack EPUB archives, detect and parse XHTML content, render each page through Puppeteer using exact viewport dimensions for fixed-layout fidelity
- Inject fonts via @font-face rules with configurable font replacement and global CSS overrides to maintain consistent text/image/print rendering
- Support batch conversion with PDF generation and merging via pdf-lib
- Implement cross-platform extraction logic and automatic cleanup of temporary artifacts

SKILLS

Languages: JavaScript, TypeScript, C++, C#, Java, Go, Python

Backend: Node.js, Express.js, NestJS, REST API Design

Databases: MongoDB, PostgreSQL, MySQL

DevOps/Cloud: Docker, Azure, Cloud Storage, CI/CD, Linux, Git

Testing: Jest, Mocha

Frontend: HTML, CSS, Bootstrap, React.js

Other: System architecture design, third-party API integration

EDUCATION

Bachelor of Computer Science – Mansoura University

2021 – 2025